Diurnal and Seasonal Variations of Meteor Velocity Observed at the Middle Latitude by the Mengcheng Meteor Radar

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Key Points:

- Azimuthal direction of the maximum meteor velocity changes clockwise, from the north (local morning) to the south (local evening)
- Meteor velocity distribution has two Gaussian peaks at ~28 km/s and ~54 km/s, revealing bi-annual and annual variations respectively
- Crucial meteor showers occur with 45–97% higher counts than the background, causing velocity peaks to increase

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Abstract

Meteor velocity is crucial in identifying key astronomical aspects of the meteoroid environment, including the influx of meteoric material and the distributions of meteoric radiants. The long-term observation and investigations of meteor velocity characteristics at mid-latitudes are scarce. This study investigates diurnal and seasonal variations of meteor velocity from April 2014 to June 2023 observed by the meteor radar at Mengcheng (33.4°N, 116.3°E). The main results are as follows. Regarding the diurnal variation due to the Earth's rotation, we found that (a) meteor velocity is faster in the morning and slower in the afternoon and evening; (b) meteor azimuth peak rotates clockwise with local time; (c) meteor zenith peak increases in the morning and decreases in the afternoon and evening; (d) the azimuthal direction of maximum meteor velocity shifting clockwise from the north in the morning to the south in the evening. In addition, the meteor velocity distribution resembles a superposition of two Gaussian distributions, with the lower (higher) distribution mainly ranging from 20 to 40 (45 to 65) km/s. Two velocity peaks are estimated using the Double-Gaussian fitting approach. In terms of seasonal variation, we found that (a) the low-velocity peak indicates bi-annual variation, with maxima in June-July and December-January, and minima in February and August; (b) the high-velocity peak indicates annual variation, with maxima in September-December and minima in March. Statistical analysis revealed notable meteor shower events where the meteor counts exceeded 45–97% of the background count on corresponding event dates and velocities.

Keyword: meteor, meteor velocity, meteor radar, meteor shower

Plain Language Summary

Meteor velocity plays a crucial role in understanding the quantity and distribution of meteoroids in our environment, yet there are limited observations and investigations conducted at mid-latitudes. In this study, we utilized a meteor radar located in Mengcheng (at the northern middle latitude) to investigate diurnal and seasonal variations in meteor velocity from April 2014 to June 2023. Our findings revealed that meteor velocity is generally faster in the morning and slower in the afternoon and evening, and the angle of arrival of meteors varies with local time. Additionally, we discovered that meteor velocity exhibits two peaks, with the lower one at about 28 km/s and the higher one at 54 km/s. Seasonal changes indicated a bi-annual variation in the low-velocity peak and an annual variation in the high-velocity peak. Furthermore, we identified several noteworthy meteor showers.

1 Introduction

Meteoroids are the small particles that enter the Earth's atmosphere at high velocities, creating plasma density enhancement trails detectable by radar as radio meteors and also optically as ‘meteors’ when they generate light. They typically occur at altitudes below 130 km where the air is dense enough to burn and ionize their surface layers (Ceplecha et al., 1998; Hughes, 2020). Once formed, the ionized trails drift with the local wind and diffuse into the background atmosphere. The diffused meteor material, which contains some metal atoms, precipitates into the atmosphere and is moved downward. It has been quite challenging to determine with accuracy just how much meteoric material enters the Earth's atmosphere. Older values range between 5 and 300 tonnes per day (Brown et al., 2002; Ceplecha, 1992). More recently, Drolshagen et al. (2017) estimate that 54 tonnes of extraterrestrial debris (interplanetary dust, meteoroids, or asteroids) fall to Earth every day, while Carrillo-Sánchez et al. (2020) estimate daily values of meteoroid flux

from 15 to 60 tonnes. However, what is clear is that the highest mass influx is caused by meteoroids with sizes between a few tens of microns and one millimeter.

The influx of meteoric material affects the chemistry of the atmosphere, the production of aerosols, and the productivity of the oceans, which has prompted more studies to comprehend the variation of the influx of interplanetary dust (Rudraswami et al., 2021). Meteoric deposition also results in layers of free neutral metal atoms in the mesosphere/lower thermosphere (MLT) (~80 to 110 km), which are readily detected using ground-based lidars (see e. g., Bowman et al., 1969). The vaporized material due to meteors also leads to the formation of sporadic E layers, which consist of metallic ions at ionospheric E region heights, and are particularly important for radio-wave propagation (see e.g., Murayama et al., 2008). Meteor plasma trails can also generate plasma instabilities in the ionosphere (Dou et al., 2010; Kelley, 2004; Oppenheim & Dimant, 2006). As a result, studying the meteoroid flux and its seasonal change is crucial for advancing our understanding of its impact on our natural environment.

Radar has been used to routinely detect meteor trails since shortly after the second world war (see e.g., Reid and Younger, 2016). Fainter meteors that result from meteoroids in the sub-mm size range are typically detectable by radar surveys. The sort of echo that different radars can detect (meteor head echo vs. trail echo, for example) and the size range depend heavily on the transmitted power and system aperture (see e. g., Janches et al., 2014, 2015). Large data sets are typically produced because the number of meteoroids identified by a single radar station far outnumbers the more infrequently detected optical meteors. Moreover, radar surveys can run continuously throughout the day, allowing for the observation of meteors coming from the Helion source. Meteor radars have some advantages that other sensor systems do not have. As long as there are ionized meteor trails available for observation, meteor radars may function day or night, in any weather, unlike optical observations, rockets, and satellites. Because of this, meteor radars may be used to detect a continual stream of cosmic debris regardless of the time of day or the weather (Marino et al., 2022; Younger, 2011).

Drolshagen et al. (2020) review many previous studies of meteoroid velocity distributions, including those derived from the Harvard Radio Meteor Project, 1968-69 Synoptic Year Program. The most complete of these is by Taylor and Elford (1998) who reanalyzed the orbital element distributions of the detected meteoroids to correct for several potential biases including the observation times, the radar antenna beam patterns, and the radar detectability of the meteoroid and the characteristics of the atmosphere. They presented the resulting radiant distribution for meteoroids with masses greater than 10^{-4} g encountering the Earth. This shows the six radiant sources as the Helion, Anti-Helion, North Toroidal, South Toroidal, North Apex, and South Apex. Subsequent work by Galligan and Baggaley (2004) using the Advanced Meteor Orbit Radar found similar orbital distributions but slightly larger velocities. We note that Drolshagen et al. (2020) point out that both optical and radar studies of meteoroids are biased against smaller slower meteoroids and go on to focus on meteoroids larger than 1 g in their work.

To accurately determine the orbit of meteor streams, it is crucial to have precise knowledge of the velocity of shower meteors toward Earth (Younger et al., 2012). Holdsworth et al. (2008) analyzed the annual variation of meteor velocity observed by the Davis meteor radar in Antarctica and found that the meteor velocity distribution peaks at around 25 km/s. The form of the distribution they found did not change much throughout the course of the year, except for a secondary peak that can be seen at around 55 km/s, with its center around early March. Singer et al. (2004) reported the meteor flux and velocity observed by the meteor radar located in Juliusruh

(54.6°N, 13.4°E) and at the ALOMAR observatory on Andøya (69.3°N, 16. 0°E) in the Arctic and found that the meteor rate in June showed a significant peak and that the meteors detected may have originated from sporadic meteoroids with large ecliptic inclinations and low to moderate atmospheric entry velocities.

A study of diurnal and seasonal variation of meteoric flux at the South Pole using a VHF meteor radar was carried out by Janches et al. (2004), who found that as there is an asymmetry in the flux recorded at both poles, it is likely that the majority of meteoric activity is centered on the ecliptic plane and is more observable during the polar summer season. Younger et al. (2009) investigated the distribution of meteor sporadic radiant observed by meteor radars in the Arctic, Antarctic, and near the Equator, and found that the observed meteor count rates showed dependence on latitude and time. This dependence is likely due to multiple factors, including the distribution of sporadic radiant, yearly fluctuations in atmospheric density, and the radar's sensitivity towards meteors originating from varying source elevations, in addition to the background temporal and spatial changes in meteor flux. Li et al. (2022) reported short-duration meteor showers at Mohe (53.49°N), Beijing (40.3°N), Wuhan (30.54°N), and Ledong (18.44°N) in China. However, the knowledge of meteor velocity and their seasonal variation characteristics at midlatitudes is sparse.

Therefore, more than nine years of observations using the Mengcheng meteor radar is a good supplement for meteoric observation at mid-latitudes. The goal of this study is to introduce meteor velocity observation by the University of Science and Technology of China (USTC) meteor radar at Mengcheng station. This paper is structured as follows. In Section 2, we introduce the observational instruments and datasets. In Section 3, we present the diurnal variations of meteor velocity and angle of arrival (azimuth and zenith). In Section 4, we perform an analysis of the seasonal variation of the meteor velocity's Double-Gaussian distribution. Lastly, we offer a summary of our findings and draw our conclusions.

2 Instruments and Datasets

The velocity distributions of meteors and their seasonal variation are investigated in this study using data obtained from the USTC Mengcheng meteor radar (MCMR) from April 2014 to June 2023. The USTC MCMR, which is part of the ATRAD Enhanced Meteor Detection Radar (EMDR) series, is located in the northern middle latitudes of China and is specifically situated in Mengcheng (33.36°N, 116.49°E). It is analogous to the Buckland Park meteor radar system reported by Holdsworth et al. (2004). The radar, which was put into operation in April 2014, has been working with 24 kW for transmitting. Five two-element Yagi antennas composing a cross '+' shape arrangement are used for reception. By examining the phase drift of echoes from sparse meteor trails, the radar is primarily used to determine winds in the MLT region. Table 1 provides a summary of the main operational characteristics of the MCMR.

By employing simple interferometry, signals captured from several antenna pairs were auto- and cross-correlated to determine the radial velocity, altitude, zenith, and azimuth angles for each meteor echo (Jones et al., 1998). The information to detect the meteor's position in the sky with an angular precision of higher than 2° and a range resolution of 1.8 km is provided by interferometric analysis of the received meteor signal.

Table 1. The main operational characteristics of the Mengcheng meteor radar.

Frequency	38.9 MHz

Peak power	24 kW at the transmitter
Pulse repetition frequency	430 Hz
Transmission mode	Circular
Coherent integrations	4
Range resolution	1.8 km
Pulse type	Gaussian
Pulse width	24 μs
Duty cycle	15%
Detection range	70–110 km

A basic overview of the meteor echoes observed by the MCMR is shown in Figure 1. As shown in Figure 1a, which displays the horizontal projection of 12337 meteors recorded by MCMR on October 2, 2022, meteors are detected in all horizontal directions and mostly in the daytime (see Figure 1e for color scale). The backscattered echoes are seen in a 300 km radius around the MCMR, most of which are located between 50 and 150 km away. Figure 1b presents the local time distribution of meteor echoes observed on October 2, 2022, by the MCMR. As evident from the figure, the meteor count rate exhibits a distinct diurnal variation, higher during the morning (i.e., 0–12 local time, or LT for short) and lower from the local afternoon until midnight (i.e., 12–24 LT). The zenith angles are predominantly dispersed between 45° and 70° and up to 80 from the zenith (Figure 1c), which shows good accordance with Singer et al. (2004), who indicated that a considerable portion of the meteor trail echoes at elevation angles at 35° or less. According to Figure 1d, the meteor echoes are observed within an altitude range of around 70 and 110 km, with a peak height of 90. 00 km and a width of 5.575 km. The neutral density affects the meteor distribution's peak height, which can fluctuate seasonally by a few kilometers which will be presented in Figure 2. The azimuth distribution of meteors is shown in Figure 1e. The azimuth angles are rather well distributed, with a somewhat higher concentration to the north of the receivers (i.e., 350°–20°).

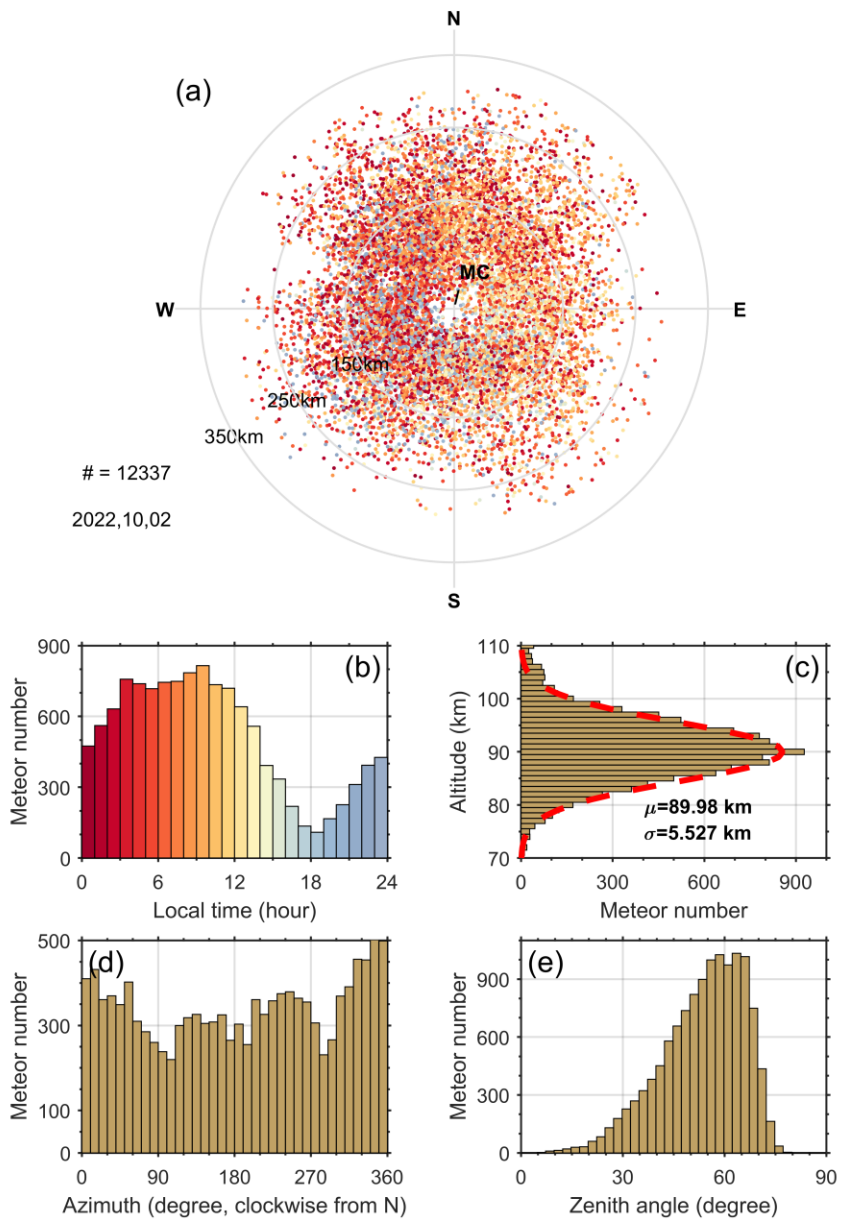

Figure 1. The meteor distribution map for October 2, 2022, is shown in panel (a), and the color of each echo is denoted by its corresponding detection in local time, as shown in the color scale in panel (b). The (c) altitude, (d) zenith, and (e) azimuth of meteor echo distributions were observed by the MCMR on October 2, 2022.

Figure 2 displays the seasonal variation of three parameters measured by the MCMR from April 2014 to June 2023: (a) the daily count rate, and (b) the peak and width of meteor height distribution on October 2, 2022. In this study, days with a meteor count below 1000 were excluded to minimize possible errors associated with low meteor counts. Although there are minor data gaps mainly due to site maintenance, the MCMR has consistently exhibited a high ability to detect meteor velocities at lower mid-latitudes.

Figure 2a displays the seasonal variation in the meteor count rates observed by the MCMR at the northern lower midlatitude, which exhibits a significant seasonal variation, with a maximum in mid-autumn, relatively stable values from January to April, and a minimum in late May. Since the radar commenced gathering data, the daily meteor detection rate has typically ranged from 6,000 to 20,000. Changes in the observation location of the main sporadic meteor sources in space seasonally are primarily responsible for this variation (Campbellbrown, 2008).

The meteor count rate basically shows annual variation in the Northern and Southern Hemispheres, except for the season of peak meteor count rate varies with latitudes. For example, at Davis Station (68.6°S, 77.9°E) in Antarctica, meteor count rates show a maximum in January and a minimum in August (Yi et al., 2017). Also, at Svalbard (78.3°N, 16°E) and Tromsø (69.6°N, 19.2°E) meteor radar located in the Arctic, count rates show a maximum in July and a minimum in February (Yi et al., 2021). Singer et al. (2004) reported that count rates at Juliusruh (54.6°N, 13.4°E) and ALOMAR (69.3°N, 16. 0°E) both show the largest values in June and the lowest count rates in February. Additionally, meteor radar in the northern middle latitudes, specifically Mohe (53.5°N, 122.3°E) and Beijing (40.3°N, 116.2°E), presented relatively weak seasonal variations with maxima in September and October and minima in February and March, respectively (Yi et al., 2021). Yi et al. (2018) reported that the meteor count rates observed by the Kunming meteor radar at northern low latitudes (25.6°N, 108.3°E) exhibit annual variation, peaking in July and minima in late March. However, Janches et al. (2004) and Reid et al. (2006) found that meteor count rates at southern high latitudes peaked in January and peaked in June. The Earth's orbital position in relation to sporadic meteor sources, the radar's sensitivity to meteors from various source altitudes, and the location of the observation site all contribute to the annual variance of the count rates (Younger et al., 2009).

In Figure 2b, the peak height observed by the MCMR shows annual variation, with a maximum value approximate of 92 km in winter and a minimum value approximate of 88.5 km in summer. The width of meteor distribution fluctuates from 4.5 km in summer to 6.5 km in winter. The peak height of meteor distribution can be used to indicate variations in mesospheric density, but it is also significantly affected by seasonal variations in meteor velocity (e.g., Yi et al., 2017, 2018). The peak height in Davis Station (68.6°S) mainly shows annual variation, reaching its maximum in summer (January) and minimum in winter (July). Similarly, the meteor radar located in the Arctic stations of Svalbard (78.3°N) and Tromsø (69.6°N) displays peak heights that primarily follow an annual variation with an additional semiannual variation. The maximum peak height occurs in autumn (October) while the minimum height is observed in spring (February). The peak altitude at Mohe (53.5°N) (higher northern midlatitudes) indicates a semiannual oscillation and a relatively weak annual oscillation. At northern mid-latitudes, the peak height in Beijing (40.3°N) mainly exhibits annual variations, with the highest peak height recorded in autumn (October-November) and the minimum during summer (July) (e. g, Liu et al., 2016; Yi et al., 2021).

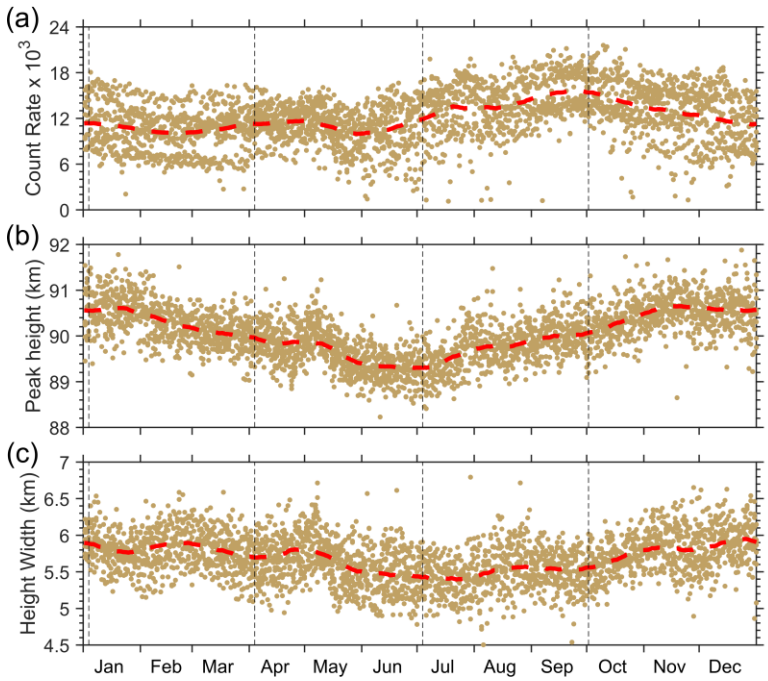

Figure 2. Annual composition of (a) the daily count rate, (b) the peak, and (c) the width of height distribution of meteor echoes observed by the MCMR from April 2014 to June 2023. The red line denotes the 30-days smoothed mean value of the daily count rate, while the black dashed lines denote the four dates displayed in Figure 3.

Figure 3 is a schematic illustrating the viewing geometry of the MCMR. The points highlighted are perihelion (January 4, 2022), aphelion (July 4, 2022), and the equinoxes between them (April 4, 2022, and October 2, 2022, in spring and autumn, respectively). In section 3, we primarily concentrate on these four positions in the Earth's orbit to analyze the variation of meteoroid fluxes along the Earth's route about the Sun.

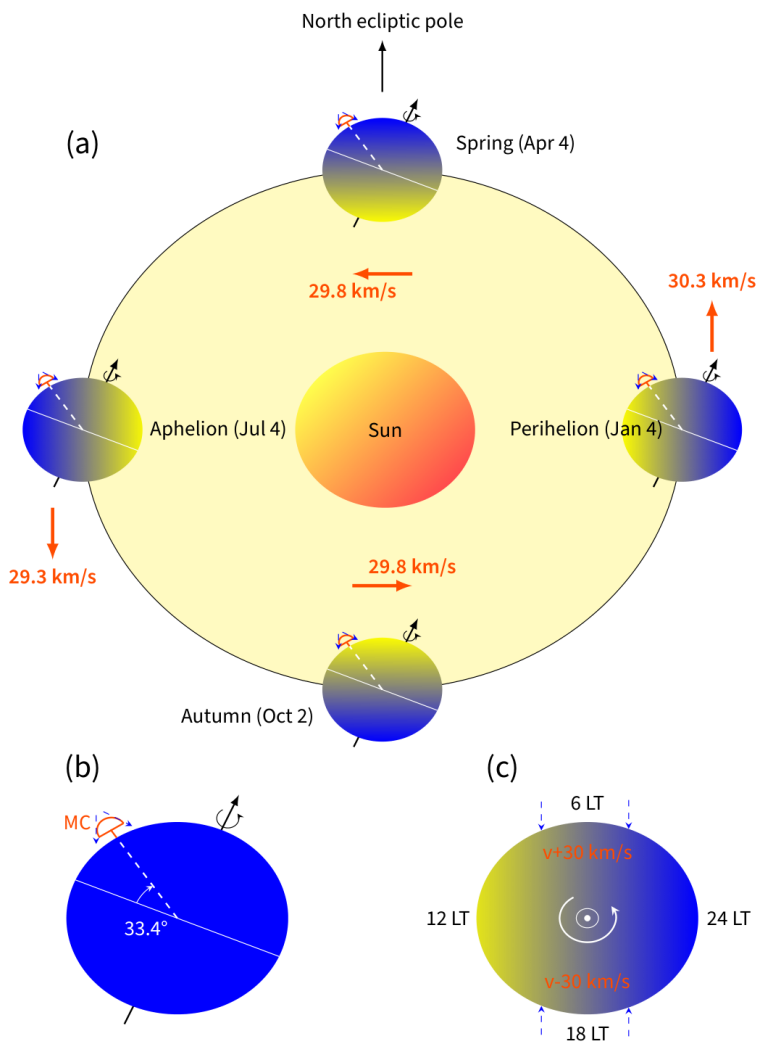

Figure 3. Schematic illustrating the viewing geometry of the MCMR. (a) the position of MCMR in the solar system changes as a result of Earth's rotation and Earth's orbit around the Sun; (b) the location of MCMR on Earth; (c) the viewing geometry of MCMR varies with local time, causing variation in the observed meteor velocity and angle of arrival, see down from the North Ecliptic Pole.

3 Diurnal variations of meteor velocity and angle of arrival (azimuth and zenith)

Figure 4a shows the diurnal variation of the meteor azimuth observed by the MCMR during October 1–3, 2022. The data is processed using a running mean with a 3-hour window to provide a minimal amount of smoothing. Figures 4b-4e show the 30-day composite diurnal variation of meteor azimuth centered on four selected dates. Figure 5 shows the diurnal variation of the meteor zenith observed by the MCMR. As shown in Figure 5a, the zenith distributions generally reach their maxima at 60°–70°. In Figures 5b-5e, the zenith angles observed by the MCMR are generally stable at about 60°–70°, except at 14–18 LT, the zenith angle decreases significantly to 45°.

In general, the change in the azimuth direction with the highest number of meteors (or azimuth peak) occurs in a clockwise manner. Specifically, the direction of the azimuth peak changes from the northwest (270–360°) to the northeast (0–90°) during the local morning (0–12

LT), while during the local afternoon to evening (12–24 LT), the direction of azimuth peak changes from the southeast (90–270°) to the southwest (180–270°). On the other hand, the meteor zenith angle also exhibits a strong diurnal variation. More precisely, the zenith direction with the highest number of meteors (or zenith peak) increases during the local morning (0–12 LT) but decreases during the local afternoon to evening (12–24 LT). Additionally, while the meteor zenith demonstrates a concentrated distribution between 0–12 LT, it tends to display a more dispersed distribution during 12–24 LT, reaching its utmost scatter around 18 LT.

From Figures 4b-4e and 5b-5e, these diurnal variations of meteor angle of arrival (azimuth and zenith) are consistent throughout the year and are primarily attributed to the counterclockwise rotation of the Earth (see Figure 3c). The meteor radar's direction towards the primary source region of meteors varies with the rotation of the Earth, which has a period of one day, causing the azimuth peak to change in a clockwise manner while causing the zenith peak to increase in the morning and decrease in the afternoon and evening. These results are consistent with Lau et al. 's (2006) model of Earth's rotation.

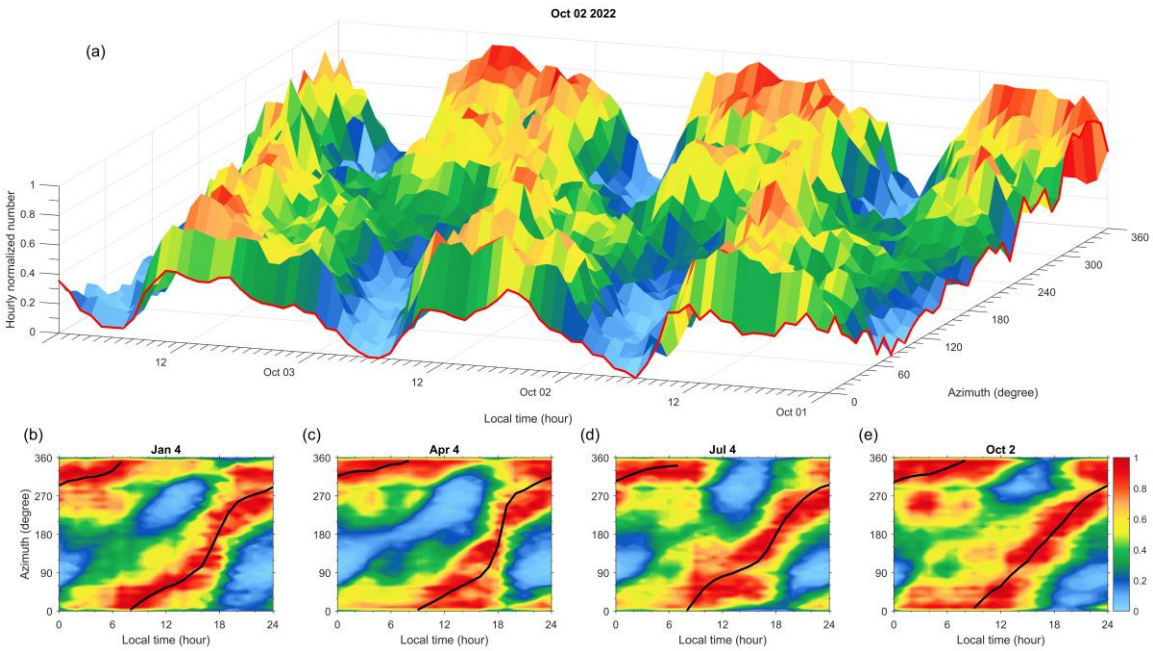

Figure 4. Panel (a) shows the diurnal variation of meteor azimuth distribution from October 1 to October 3, 2022 (a running mean with a 3-hour window), where the red lines around the figure box denote the local time distribution of meteors on the 0 azimuths, and the azimuth distribution of meteors on the 0 local times, respectively. Panel (b)-(e) presents the diurnal variation of meteor azimuth distribution, 30-day composition centered on January 4, April 4, July 4, and October 2, 2022, respectively. The number densities of all panels have been hourly normalized to the maximum number density. The black lines indicate variations in the azimuth peak corresponding to local time.

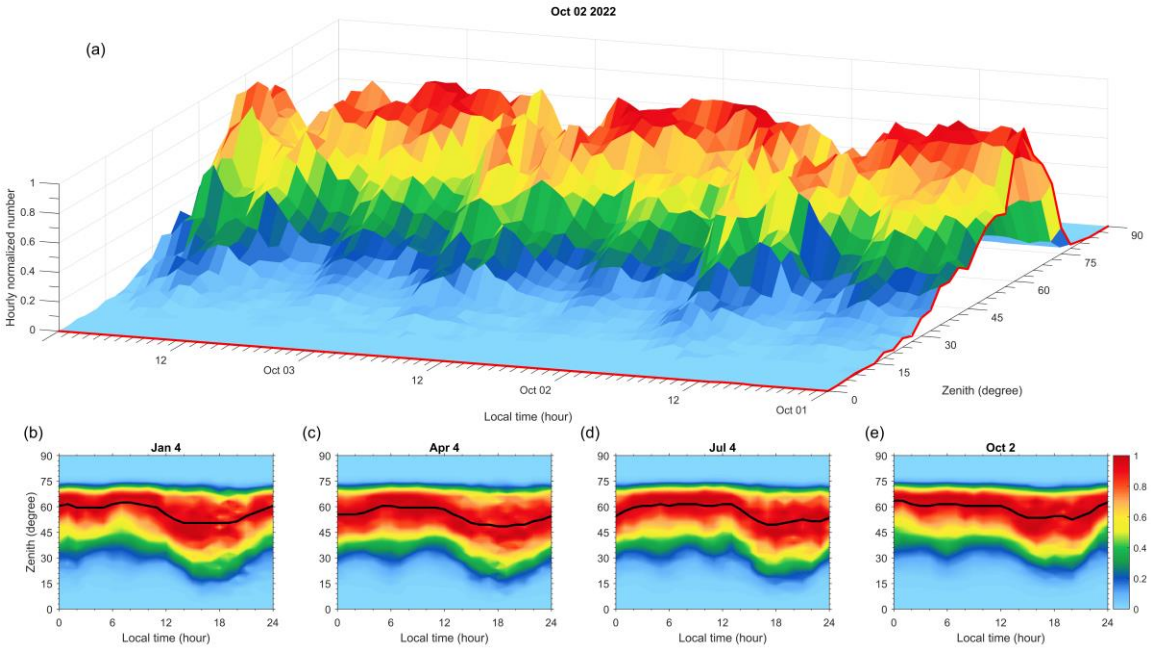

Figure 5. Same as Figure 4, but for meteor zenith angle.

Figure 6 illustrates the diurnal variations in meteor height and velocity over a period of 30 days. The MCMR records a distinct diurnal variation in meteor velocity, showing higher velocity in the local morning (0–12 LT) and lower velocity from the local afternoon to midnight (12–24 LT). This phenomenon occurs due to the Earth's rotation (see Figure 3c) and is well-explained by Sugar (1964): On the morning side of Earth, meteors are swept up by Earth's forward motion as it orbits the Sun. Conversely, on the evening side, only meteors that overtake the planet are able to reach it. This uneven distribution leads to a peak occurrence rate and maximum velocity at approximately 6 LT and a trough in occurrence rate and minimum velocity at around 18 LT. The degree of variation in this ratio is contingent on the observer's latitude. Kero et al. (2012) reported that the diurnal detection rate exhibits an asymmetrical seasonal variation near the equinoxes. The months preceding the Spring equinox (around March 20) exhibit higher daily maxima and overall diurnal numbers, compared to the subsequent months.

Meteors travel at different velocities as they enter the Earth's atmosphere. These velocities determine the altitude ranges in which they can be detected. Rapid-moving meteors quickly burn up into plasma, leaving a detectable trail at higher altitudes. Slow-moving meteors burn up more gradually and can only be detected at lower altitudes. Thus, meteor altitude decreases at 18 LT following a decrease in meteor velocity. The maximum meteor altitudes are found with the fastest meteors, whereas the slowest meteors produce lower altitudes.

In addition, Figure 6 reveals that there is a seasonal variation in meteor height and velocity. The peak height on four selected days generally followed the results shown in Figure 2b, which indicates higher peak height in winter and lower peak height in summer, which can be referenced by Yi et al. (2019). Moreover, Figures 6e-6h identified January 4 and July 4 to be when there are the largest and smallest changes in meteor velocity, respectively. It is important to acknowledge that the diurnal variations of atmospheric density can play a significant role in changes in meteor altitude. It can be complex to distinguish the effects of atmospheric density and meteor velocity in

such changes. While the acceleration of a meteor can provide insight into changes in atmospheric density, it is not within the scope of this study and will be investigated further in future work.

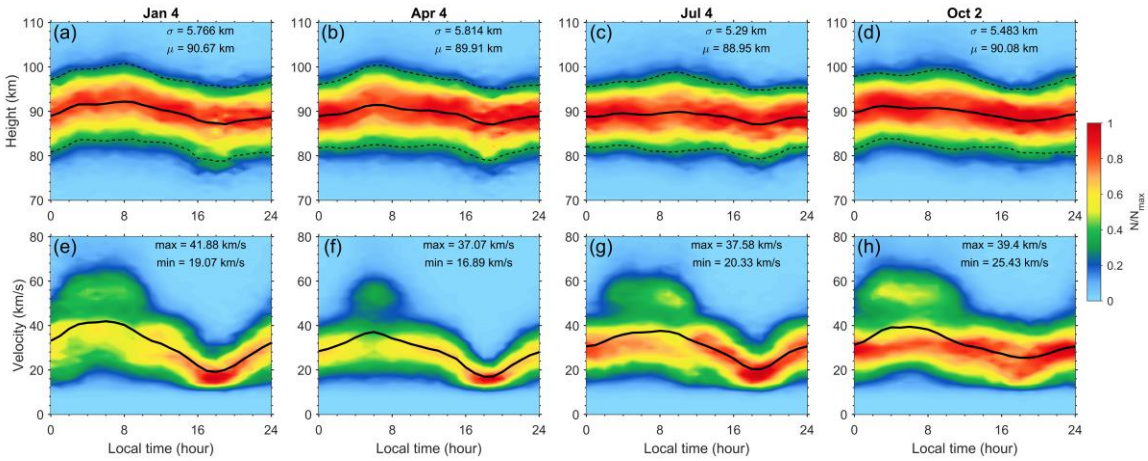

Figure 6. The 30-day composition of diurnal meteor height and velocity variations centered on January 4 (column 1), April 4 (column 2), July 4 (column 3), and October 2, 2022 (column 4), respectively. Top row (a)-(d): Diurnal variations of meteor height. The black lines and black dashed lines denote the peak and width of meteor height distribution in local hours. The peak and width values of daily meteor observation are labeled on the top right. Bottom row (i)-(l): Diurnal variations of meteor velocity. The black lines show the mean value of meteor velocities every hour, and their maximum and minimum are labeled on the top right. The number densities of all panels have been hourly normalized to the maximum number density.

These findings imply that there exist variations in the azimuth, range, altitude, and velocity of meteors. To better examine the relationship between meteor azimuth and meteor velocity, the collected data was partitioned into 3-hour intervals based on local time (ranging from 0 LT–3 LT to 21 LT –24 LT), and subsequently plotted in Figure 7. Notably, the direction of the maximum velocity exhibited a clockwise rotation with increasing time. To quantify this directional tendency, we computed a vector, denoted as $\boldsymbol{Q}$, using Equation (1):

$$\boldsymbol{Q}(t_1, t_2) = < v_i \cos \phi_i > \boldsymbol{e_x} + < v_i \sin \phi_i > \boldsymbol{e_y} \quad (1)$$

where v_i and ϕ_i denote the velocity and azimuth angle of the i-th meteor, respectively, and <> denotes an ensemble average. A vector is formed by this average, pointing towards the general direction of the higher values of v, indicating the direction of the quickest meteor velocity. These vectors are illustrated in Figure 7, positioned at the center of each diagram, with a scaling factor of 53 (i.e., the length of each vector of 1 km/s is equivalent to 53 km on the map).

In general, there is a clockwise occurrence of changes in the direction of the maximum meteor velocity. Specifically, during the local morning (0–12 LT), the direction changes from the northwest (270°–360°) to the northeast (0°–90°), while during the local afternoon to evening (12–24 LT), the direction of the maximum meteor velocity changes from the southeast (90°–270°) to the southwest (180°–270°). This phenomenon is similar to the change in the azimuth direction of the maximum meteor number and can be attributed to the self-rotation of the Earth. At sunrise, the fastest meteors originate from the north, which can be explained by the location of the MCMR in the Northern Hemisphere. On the other hand, meteors are the slowest at sunset. The meteor radar's

changing direction, as the Earth rotates, relative to the local time, is responsible for the longitude difference from the sunrise. Consequently, the highest number of fast-moving meteors is observed coming from the north during the morning, while the opposite effect occurs in the evening.

However, the physical interpretation of the magnitude of $\boldsymbol{Q}$ is not readily apparent, as larger values of average velocity do not necessarily correspond to larger magnitudes of $\boldsymbol{Q}$. This is because $\boldsymbol{Q}$ represents the average value of the velocity vector concerning the azimuth angle, and the magnitude of $\boldsymbol{Q}$ is influenced by both the velocity and azimuth angle distributions of the meteoroids. Specifically, when the distribution of azimuth angles for meteoroids is highly uniform, the magnitude of $\boldsymbol{Q}$ becomes smaller (i.e., $\boldsymbol{Q}$ equals zero when meteoroid velocities are uniformly distributed across all azimuth directions). Conversely, when the distribution of azimuth angles is highly non-uniform, the magnitude of $\boldsymbol{Q}$ becomes larger. Furthermore, a larger overall velocity of the meteoroids corresponds to a larger magnitude of $\boldsymbol{Q}$. Hence, the value of $\boldsymbol{Q}$ may surpass during the time interval between 21 LT to 24 LT compared to that in the interval from 6 LT to 9 LT, despite the meteor velocity being relatively lower in the local evening as compared to the velocity in the local morning.

The meteor velocity distributions depicted in Figures 6 and 7 are highly consistent. Sugar (1964) states that the velocities of meteors measured by meteor radars during the daytime are added by Earth's orbital velocity around the Sun (approximately 30 km/s), whereas during the nighttime, the orbital velocity is deducted. Therefore, during the daytime, both high-velocity and low-velocity meteors can be observed, leading to a two-peak distribution. Nevertheless, during the nighttime, high-velocity meteors experience retardation due to the Earth's orbital velocity around the Sun, resulting in a combination with the low-velocity meteors in the velocity distribution plot, so the velocity distribution during the nighttime shows only a single peak.

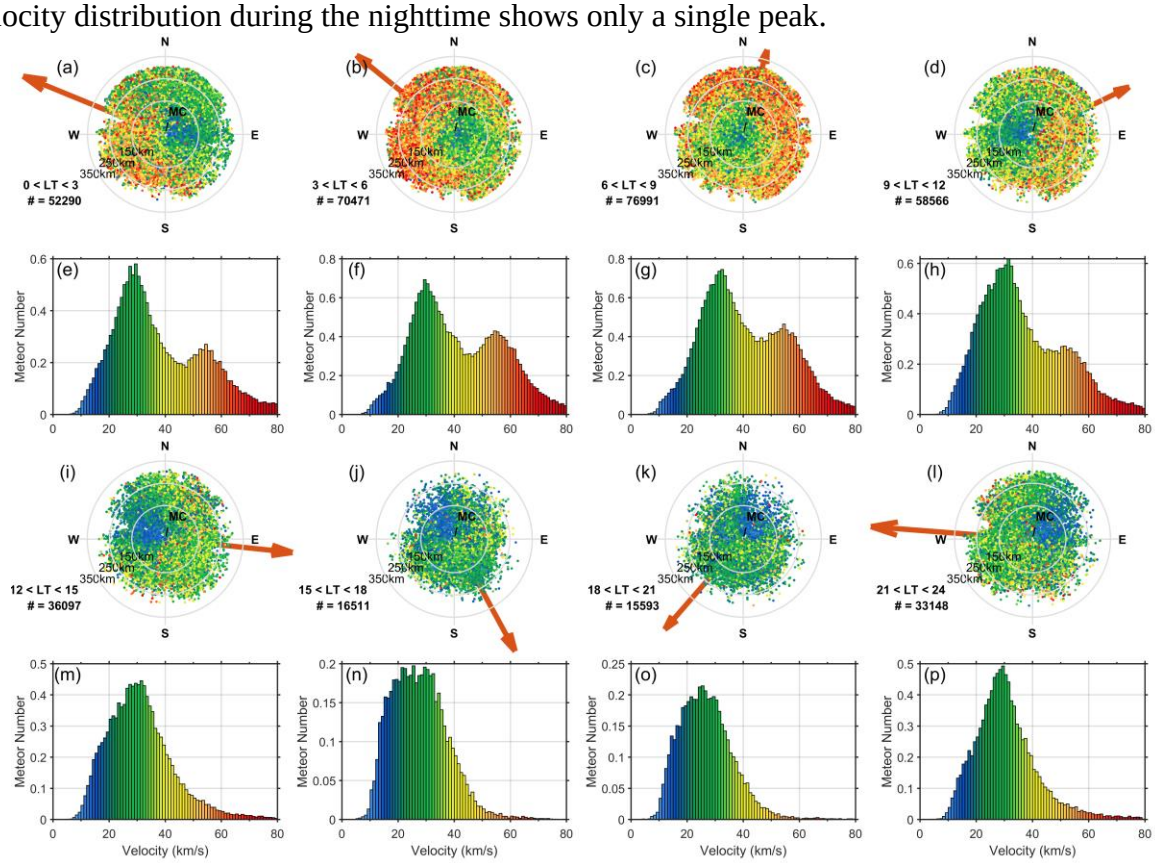

Figure 7. The first and third rows depict distribution maps of meteor velocities as a function of horizontal distance and azimuth, which are sorted into 8 local time intervals over the composition of a 30-day period centered on October 2, 2022. Each meteor is represented as a colored point, with different colors used to indicate its velocity, as shown in the second and fourth rows where color scales are provided. Notably, " LT" is an abbreviation for Mengcheng local time (approximately UT+8 h). The circles in maps in the first and third rows denote distances of 150 km, 250 km, and 350 km.

4 Seasonal variations of the meteor velocity

Figure 8 shows the meteor range-velocity distribution, meteor altitude-velocity distribution, and meteor velocity distribution for the past 30 days. The distributions of velocities and altitudes reveal a pronounced relationship between meteor velocity and altitude. According to meteor physics, meteors that are traveling quickly start to burn at high altitudes that are visible on the charts. According to the findings of other studies, there has been a localized rise in the number of meteors traveling at velocities of around 25 km/s and 55 km/s (Holdsworth et al., 2008). This characteristic of the velocity distribution may be linked to the heterogeneity in meteoric material dispersion close to the Earth's orbit. Comparing the mutual distribution of meteors by velocity and radiant is important to provide a more conclusive response to this query. It might suggest the existence of solitary meteoroid streams if close radiants exhibit high-velocity meteors traveling at velocities of up to 55 km/s (Kalabanov et al., 2021).

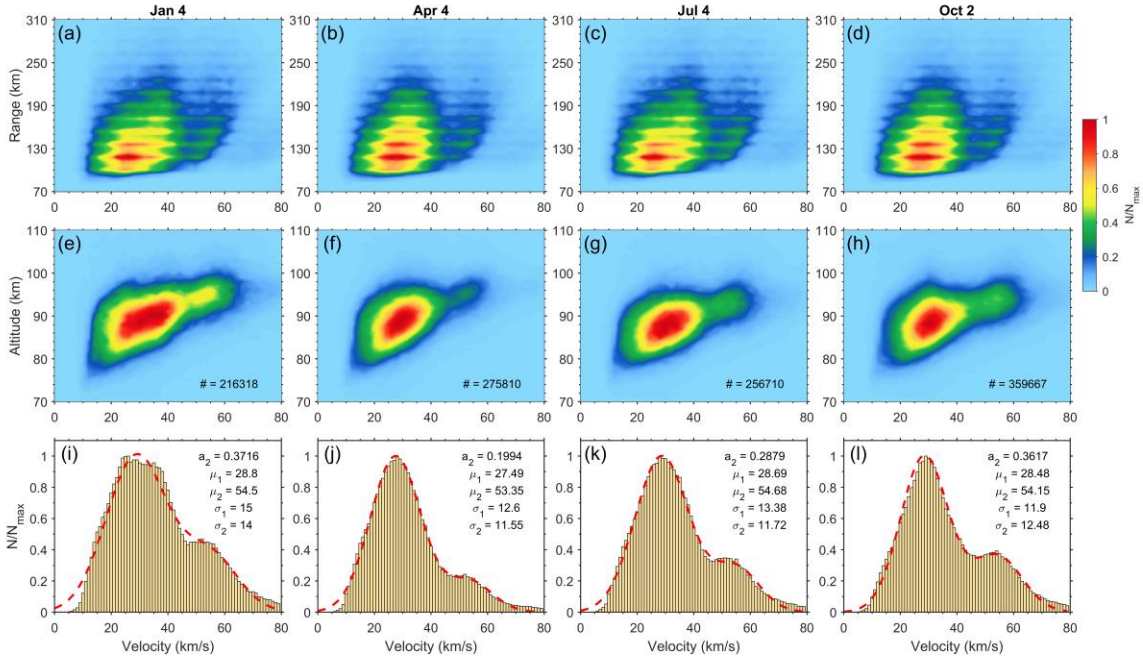

Figure 8. The 30-day composition of meteor velocity distribution centered on January 4 (column 1), April 4 (column 2), July 4 (column 3), and October 2, 2022 (column 4), respectively. Top row (a)-(d): meteor range-velocity distribution. Second row (e)-(h): meteor altitude-velocity distribution. The number densities of (a)-(h) have been normalized to the maximum number, which is shown on the top-right of the second row (e)-(h). Bottom row (i)-(l): histograms and Double-Gaussian fitting curves for meteor velocity distribution centered on the (a) January 4, (b) April 4,

(c) July 4, and (d) October 2, 2022, respectively. Calculated mean and standard deviation values for each fitted Double-Gaussian curve are shown.

The bottom row in Figure 8 displays histograms of meteor composition over the course of 30 days, revealing a meteor velocity distribution that resembles a superposition of roughly two Gaussian distributions. To determine the parameters of this distribution, we utilize the Double-Gaussian fitting approach on the meteor distribution that has been normalized by its maximum count. As a result, the meteor velocity distribution may be captured by describing the velocity peaks and their corresponding widths as a function:

$$N_{\mathrm{nor}}(t) = a_1 \mathrm{e}^{-\frac{(v-\mu_1)^2}{\sigma_1^2}} + a_2 \mathrm{e}^{-\frac{(v-\mu_2)^2}{\sigma_2^2}} \tag{2}$$

where a_1 and a_2 denote the amplitude of the lower and higher velocity peak, μ_1 and μ_2 the lower and higher velocity peak, and σ_1 and σ_2 the lower and higher velocity width, respectively. Note that the velocity peaks and widths are calculated using this function where a_1 is normalized to 1 to streamline calculations.

The conclusion drawn from an analysis of the bottom row of Figure 8 is that a significant seasonal variation is present in the higher velocity peak, which ranges from around 52.6 km/s to about 54.7 km/s. Similarly, the lower velocity peak exhibits a difference in various seasons and ranges between approximately 27 km/s to 28 km/s. The changes in high and lower velocity peaks were measured by the ratio $R = a_2/a_1$, which reveals that in comparison to the higher velocity peaks, the lower velocity peaks are more apparent during the Northern Hemisphere autumn and winter than in spring and summer (i.e., R is close to 0.36 near January 4 and October 2, whereas it is close to 0.25 near April 4 and July 4).

Following Equation (2), we utilize the Double-Gaussian fitting approach to exhibit the seasonal changes in meteor velocity distributions, including the daily high and lower velocity peaks. Figure 9a displays a comprehensive analysis of 39.6 million meteor echoes captured from April 2014 to June 2023. The nine-year mean seasonal variation of the daily meteor velocity distributions is shown in Figure 9b. In general, meteors with velocities between 20 and 40 km/s dominate throughout the year. From August to December, there is a peak region with a velocity of about 55 km/s, which corresponds to a period of highly active meteor showers, as evident by Figure 2a.

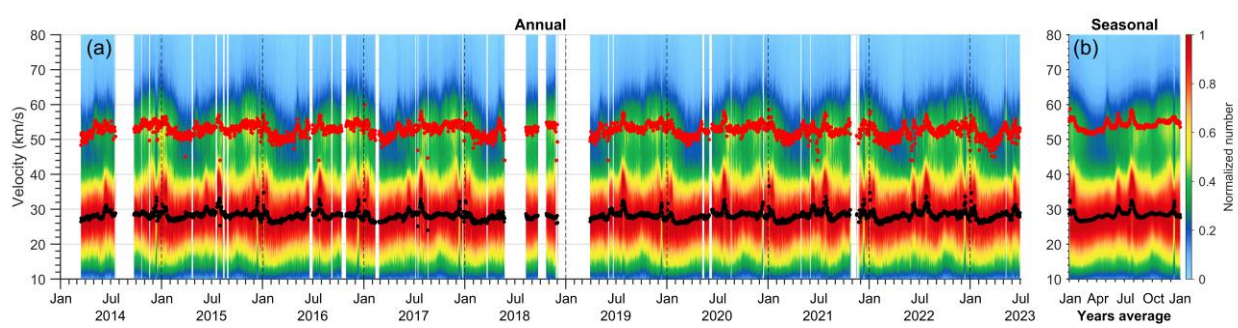

Figure 9. Panel (a) shows the nine-year variations of daily meteor velocity distributions, in which red and black dots each denote the daily lower velocity peak (μ_1) and the higher velocity peak (μ_2) observed by MCMR from April 2014 to June 2023. The color bar denotes the daily normalized

number density. Panel (b) is the composite average of nine-year velocity distributions for Panel (a).

Figures 8 and 9 provide valuable insight into the two Gaussian distributions. The lower meteor velocity Gaussian distribution ranges from 20 to 40 km/s, while the higher one ranges from 45 to 65 km/s as depicted in these figures. Seasonal variations are evident from the number densities of both distributions, as shown in Figure 9. The lower Gaussian distribution has a bi-annual variation displaying a comparatively concentrated distribution in spring and autumn, and a more scattered distribution in summer and winter. We have identified several velocity ranges between 35 km/s to 40 km/s during summer and winter, which correspond to meteor shower events and will be further discussed in Figure 10. Moreover, the higher Gaussian distribution indicates significant annual variation, which is more prominent in spring and winter and weaker in summer and autumn. Consequently, the value of a_2 is proportionately greater in Figures 8i and 8l as compared to Figures 8j and 8k, as the variance of the lower Gaussian distribution contrasts with that of the higher Gaussian distribution.

Apart from the two Gaussian distributions, Figure 9 also displays the seasonal dependence of the two Gaussian peaks at approximately 28 km/s and 55 km/s. The lower peak, indicating bi-annual variation, is around 28 km/s, while the higher peak reflects annual variation and is around 54 km/s. The minimum for the lower peak occurs around February and August, while the maximum appears in June-July and December-January. In contrast, the higher peak reaches its maximum around May-December-January and its minimum around March.

These findings are consistent with the Davis meteor radar research indicating that lower and higher peaks occur around 25 km/s and 55 km/s, respectively (Holdsworth et al., 2008), and the King Sejong Station meteor radar in 2017 demonstrating that peaks occur approximately at 30 km/s and 50 km/s (Lee et al., 2022). McNamara et al. (2004) also show similar findings employing a sporadic meteor environment model and verifying it against radar data. In their study, the lower velocity peak is about 20 km/s, while the higher velocity peak is about 55 km/s.

In addition to the seasonal variation of meteor velocity as shown in Figure 9b, there are also several noticeable increases in the meteor velocity distribution. To ascertain the reasons for the formation of the two Gaussian distributions and their seasonal variations, it is necessary to investigate the sporadic sources of meteors. Six primary apparent "sources" of sporadic meteor radiants have been identified throughout the years. Campbell-Brown (2007) stated that apex sources move at high velocities, ranging from nearly 70 km/s to approximately 45 km/s as a result of their heliocentric velocities adding to the Earth's orbit velocity. Conversely, the helion/anti-helion sources exhibit a wide range of velocities, differing between about 20 km/s (farthest from the apex) and 35 km/s (nearest to the apex). Unlike the helion and anti-helion sources, the north toroidal source moves at a uniform velocity of 35 km/s.

Furthermore, Campbellbrown (2008) suggested that the two velocity peaks are mostly caused by several sources of sporadic meteors. Apex sources produce higher-velocity meteoroids with velocities between 45 and 70 km/s, which typically have a smaller scale, while helion/anti-helion sources produce meteoroids with velocities between 20 and 35 km/s. Younger et al. (2009) explained that the low meteor count rates observed in March were due to poor viewing geometry caused by the 2 sporadic meteor sources being below the local horizon (the southern apex and southern prograde apex source regions). However, their analysis revealed that during June and July, multiple sources, including southern apex and northern ones, above the horizon, combined with a change in atmospheric density structure, resulted in large count rates.

As a result, the helion/anti-helion and apex sources both dominate the meteoroid velocity distribution seen by MCMR during the autumn and winter (see Figures 8a, 8d), whereas, during the spring and summer, only the helion/anti-helion sources dominate the meteoroid velocity distribution (see Figure 8b, 8c). These observations were also clearly supported by Kero et al. (2012), who illustrated the radiant distribution of meteors' geocentric velocity and radiant density of north and south apex sporadic meteor sources detected during the 2009–2010 MU radar head echo observation.

In addition to the sporadic meteor sources, meteor showers contribute to the variation in meteor velocities. Meteor showers occur when the Earth intersects with meteoroid streams that have similar velocities. Although meteoroid streams are unevenly distributed, resulting in varying shower intensities each year, showers occur annually between the same range of dates. Figure 9b demonstrates that local velocity peak maxima are predominantly observed on specific days throughout the year. We also investigated whether these occurrences are related to meteor showers by calculating relative variability, obtained by subtracting the smoothed meteor background. Consequently, we discovered that these local maxima are indeed associated with meteor showers which are indicated by the relative variability in Figure 10. Specifically, the meteor count surpasses 45-97% of the background count at the corresponding meteor shower event dates and velocities, causing the velocity peaks to increase.

Table 2 lists the major meteor showers shown and denoted by circles in Figure 10. It can be observed that there are comparatively small and narrow local maxima of velocity peaks increase during the shower periods, and the local maxima of the velocity peaks of one to several days are coinciding with the peak activity dates of the major meteor showers. The magnitudes of the two velocity peaks for the Quadrantids, Daytime Arietids, and southern Delta Aquariids, are higher than the 30-days smoothed background by approximately 2 km/s (H) & 2 km/s (L); 2.5 km/s (H) & 1.5 km/s (L); 2.5 km/s (H) & 2.5 km/s (L), respectively. However, during Eta Aquariids, only the higher peak reaches its local maximum, which is higher by approximately 2 km/s; during Geminids, only the higher peak reaches its local maximum, which is higher by approximately 2.5 km/s. On each plot, the approximation of the difference between the peak velocities of shower meteors and sporadic meteors is specified.

Figure 10 displays the distinctive characteristics of meteor activity, including shower meteors and sporadic meteors. The coherence of local maxima in the two velocity peaks implies that some shower meteors have higher velocities, producing their ionized trails at higher altitudes than the sporadic background, consistent with the study about altitude distributions of meteor showers and sporadic meteors made by Lukianova et al. (2020). Moreover, the velocity peaks of the meteors shown in Figure 10 exhibit considerably greater variations than those in Figure 8. This is due to the employment of the 30-day composition method in Figure 8, which mitigates the trend of variation.

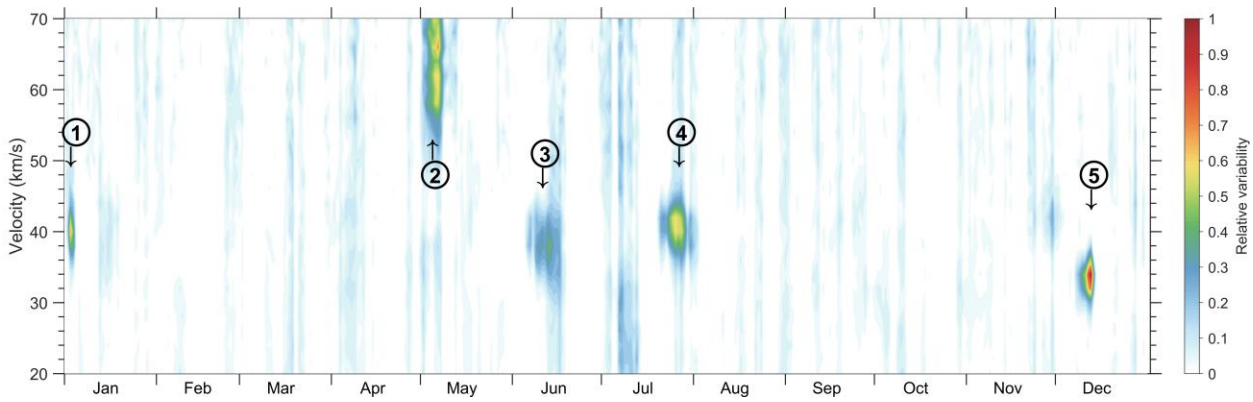

Figure 10. The seasonal variation of daily mean meteor velocity distribution anomaly for a period of more than nine years from April 2014 to June 2023. The color bar denotes the relative variability compared with the 30-day smoothed background. The numbered circles indicate the dates of meteor showers as reported by IMO (2022) and correspond to columns 4 and 5 in Table 2.

Table 2. Characteristics of meteor showers that are identifiable and numbered in Figure 10, following the shower calendar of the International Meteor Organization (IMO) for the year 2023 (IMO, 2022). The showers are listed in column 4 in order of their activity dates. Column 6 indicates the geocentric velocities (V_∞) of the meteor showers, while the final column points out the type of increased velocity peaks during the meteor shower activities (H denotes the higher peak and L denotes the lower peak).

#	Shower	IAU Code	Activity Date	Peak activity date	V_∞ (km/s)	Type
1	Quadrantids	010 QUA	28 Dec–12 Jan	04 Jan	41	H&L
2	Eta Aquariids	031 ETA	20 Apr–20 May	06 May	66	H
3	Daytime Arietids	171 ARI	14 May–24 Jun	07 Jun	38	H&L
4	Southern Delta Aquariids	005 SDA	12 Jul–23 Aug	30 Jul	41	H&L
5	Geminids	004 GEM	04 Dec–20 Dec	14 Dec	35	L

5 Conclusions

The USTC MCMR, located at (33.36°N and 116.49°E), has been gathering data on meteor count rates, range, azimuth, altitude, and velocity at the lower midlatitudes since April 2014, accumulating more than nine years of continuous observations. Our investigation yields the following salient points:

1. This study reveals that the MCMR detects approximately 6,000-20,000 underdense meteor echoes per day. Such observations yield sufficient data to determine various meteor characteristics including range, angle of arrival (zenith, azimuth), altitude, and velocity. The meteor count rates observed over Mengcheng at lower mid-latitudes of the Northern Hemisphere exhibit diurnal variations. Specifically, the meteor count rate is higher in the morning hours (2-10 LT) and lower in the afternoon through midnight hours (14-24 LT). Additionally, the meteor count rates exhibit annual variation, with the highest rate recorded from September to October and the lowest in February.

2. The diurnal variation of meteor velocity and angle of arrival (azimuth and zenith) is primarily influenced by the Earth's rotation, leading to (a) faster velocities in the morning and slower velocities in the evening; (b) meteor azimuth peak changing in a clockwise manner, varying from the north in the morning to the south in the evening; (c) the meteor zenith peak increasing in the morning and decreasing in the afternoon and evening. Analysis has also unveiled a strong correlation between velocity and azimuth, with the azimuthal direction of maximum meteor velocity shifting temporally from the north in the morning to the south in the evening. More specifically, during the local morning (0–12 LT), the azimuthal direction transitions from the northwest (270°–360°) to the northeast (0°–90°), while in the local afternoon to evening (12–24 LT), the direction of maximum meteor velocity alters from the southeast (90°–270°) to the southwest (180°–270°).

3. We found that the meteor velocity distribution can be described as a superposition of two Gaussian distributions. The lower velocity Gaussian distribution can be observed in the ~20 to ~40 km/s range, and the higher one, from ~45 to ~65 km/s. The lower Gaussian distribution demonstrates a bi-annual variation, displaying a relatively concentrated distribution in spring and autumn, and a more scattered distribution in summer and winter. The higher Gaussian distribution shows a significant annual variation, being more prominent in spring and winter and weaker in summer and autumn. The two peak velocities of the meteor velocity Gaussian distribution were estimated using a Double-Gaussian fitting approach. Generally, the peak velocity of the lower Gaussian distribution is found to be around 28 km/s and indicates a bi-annual variation. The velocity minima appear in February and August, while the maxima occur in June-July and December-January. The peak velocity of the higher Gaussian distribution, which is generally around 55 km/s, indicates an annual variation. The velocity minimizes in March, while the maxima happen in May-July and December-January.

4. By applying a superimposed statistical analysis, we were able to detect crucial meteor shower events that showed a noticeably greater number of meteors than the average background value. Specifically, the meteor counts surpassed 45-97% of the background count at the corresponding meteor shower event dates and velocities, leading to an increase in velocity peaks.

Acknowledgments

This work is supported by the National Natural Science Foundation of China (42125402 and 42174183), the National Key Technologies R&D Program of China (Grant No.2022YFF0503703), the B-type Strategic Priority Program of CAS Grant No. XDB41000000, and the foundation of the National Key Laboratory of Electromagnetic Environment and the Fundamental Research Funds for the Central Universities. WY is also funded by the Anhui Provincial Natural Science Foundation (Grant No.2008085MD113), and the Joint Open Fund of Mengcheng National Geophysical Observatory (MENGO-202209). We acknowledge the technical support of our radar systems by Chris Adami. We also express our gratitude to Jingnan Guo and Shiyuan Wang at USTC; Quanshui Zheng, Shaohua Xiang, Luping Xu, Jianxun Ren, and Liwen Jing at Shenzhen X-Institute for their valuable discussions.

Data Availability Statement

The Mengcheng meteor radar data presented in this work is archived at https://doi.org/10.5281/zenodo.8002780.

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